

Course Reflection and Learning Portfolio

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This course was one I had been genuinely excited to take. It offered a hands-on, interactive approach to learning—exactly the type of environment where I learn best. Throughout the semester, I engaged with a variety of technologies, including the Raspberry Pi, chatbot development tools, the PiCar-X, and the collaborative robot (cobot). Each module provided a different perspective on applied computing and robotics, and each experience contributed to my understanding of how hardware, software, and AI intersect.

Raspberry Pi Module

Before this class, I had heard a great deal about the Raspberry Pi but had no idea where to begin. The manual was helpful in explaining the components, schematics, and foundational concepts, but reading about the device without having it physically in front of me made it difficult to retain the programming details. By around Chapter 7, I realized that I learn far more effectively through hands-on practice than through reading code examples in isolation.

Unfortunately, due to a holiday and a weather-related cancellation, we lost two class sessions—something I felt acutely because I was eager to finally work with the device. When we eventually had the opportunity to assemble and run the Raspberry Pi, it was both exciting and challenging. The assembly process required careful attention to orientation and placement of components, and although the manual provided guidance, I found myself relying more on trial, error, and teamwork.

Once the software was installed, we experimented extensively with the code. We adjusted scroll speeds, colors, and brightness settings on the LED display, and even discovered that our initial settings were so fast we didn't realize the text was scrolling at all. That moment led to a lot of laughter—and a lot of tinkering. In our enthusiasm, we forgot to take as many pictures and video as we should have, but the experience was memorable and engaging. My only regret is that we didn't have more time to explore the Raspberry Pi further.

Chatbot Module

The chatbot assignment presented a different type of challenge. My biggest obstacle was simply getting registered on the platform. After several exchanges with technical support, I was finally able to access the system and begin experimenting. I created a few short videos, though the rendering time was much longer than expected, especially on the free plan. Even so, it was interesting to see how conversational AI tools can be used to generate multimedia content.

NVIDIA AI Conference

The NVIDIA AI conference was both informative and humbling. While some sessions were highly technical and beyond my current knowledge base, the presentations were still accessible enough for me to follow with some additional research. I frequently paused the videos to look up terminology and concepts.

One particularly memorable moment was when Jensen Huang discussed “stars.” I initially had no idea what he meant or why he was so enthusiastic. After researching, I learned that GitHub stars function as

a community-driven metric similar to product ratings. Understanding that context helped me appreciate the significance of the rapid growth he was describing.

Although some sessions were too advanced, others were engaging and helped me better understand how NVIDIA is shaping the future of graphics, AI, and computing.

AI and Robotics Presentations

The real-world applications of AI in fields such as language therapy were fascinating. Seeing how AI can assist in tracking patient progress highlighted the practical impact of these technologies. However, I was disappointed that this presentation replaced a robotics lab session. Given my preference for hands-on learning, I would have preferred that the presentation supplement rather than replace class time.

Another speaker, Intel's Jason Dunn-Potter, was particularly engaging. His ability to connect with the students and challenge conventional thinking stood out. His presentation reminded me of how many innovative ideas are being developed by students at all levels of experience. Returning to school has shown me how much there is to learn—and how many people are already working on impressive side projects. It was both inspiring and humbling.

PiCar-X Module

The PiCar-X module was, unfortunately, the most disappointing part of the course. We encountered multiple technical issues: missing cables, a non-responsive keyboard, and a weak battery that prevented the car from operating properly. While we were able to resolve some issues—such as replacing the keyboard and bringing in missing cables—the battery problem limited our ability to fully engage with the project. Our PiCar-X simply didn't have enough power to perform as expected, which made the experience frustrating.

Cobot Module

The Cobot module was the highlight of the course for me personally. I was excited to work with a collaborative robot and learn how to control its movements. Although I initially struggled with the touchpad interface and the concepts of X, Y, and Z coordinates, and still do, the instructional videos were extremely helpful. After watching them several times, I began to get the feel for how to make the Cobot move between endpoints and to do it in real life.

Once I grasped the movement logic, I was able to collaborate more effectively with my partner, Rida, who excelled at programming the Cobot. Together, we were able to plan and execute movements more confidently. Unfortunately, our third Cobot session was cancelled, which was disappointing because I felt I was just beginning to build momentum.

Overall Reflection

Despite the cancellations and technical challenges, this course was one of the most engaging and hands-on learning experiences I have had. I learned that I retain information best when I can physically interact with the technology, experiment, and troubleshoot in real time. The combination of hardware, software, AI, and robotics broadened my understanding of modern computing and reinforced my desire to continue learning in this field.